

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. (ME/EC/CL/CE/IT/EE) Examination

May 2011

ME102 Fundamental of Mechanical Engineering

Date: 13.05.2011 , Friday

Time: 1:30 p.m to 4:30 p.m

Maximum Marks: 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Draw neat sketch wherever required.

SECTION- I

- Q1(a) Classify fuel and give desirable properties of fuel. 3
- (b) Write advantages and disadvantages of non-conventional sources of energy. 4
- (c) Define heat and power, write characteristics of heat and power. 3
- Q2(a) Derive various equations for Isothermal process. 5
- (b) Determine the work done in compressing one kg of air from a volume of 0.15 m^3 at a pressure of 1.0 bar to a volume of 0.05 m^3 when the compression is (i) Isothermal and (ii) Adiabatic, take $\gamma=1.4$. Also comment on your answer. 5

OR

- Q2(a) Define Vapour, Gas and Perfect Gas. Explain with graph Boyle and Gay-Lussac law. 5
- (b) 1 Kg of air is heated at constant pressure from 27°C to 197°C . Calculate Work done, heat added and change in internal energy. If initial volume of air is 2.1 m^3 calculate pressure and final volume. Take $C_p = 1.005 \text{ KJ/kg K}$ and $C_v = 0.718 \text{ KJ/kg K}$. 5
- Q3(a) Differentiate Spark Ignition and Compression Ignition engines. 5
- (b) Explain with neat sketch working of 4 stroke petrol engine. 5
- (c) If clearance volume of otto cycle is 0.2 cm^3 and stroke volume is 1.2 cm^3 , Calculate the air standard efficiency of the engine. 5

OR

- Q3(a) Derive the Air standard efficiency of Otto cycle. 5
- (b) Explain working of 2 stroke Diesel engine. 5
- (c) A six cylinder 4 stroke engine to develop 90 KW indicated power at 800 rpm. The stroke to bore ratio is 1.5. Assume Mechanical efficiency 85 %. Brake mean effective pressure is 5 bar. Calculate cylinder diameter and stroke length. 5

SECTION II

- Q4(a) Explain steam formation at constant pressure. 3
- (b) Differentiate wet, dry saturated and superheated steam. 3
- (c) State various mountings and accessories used in Boilers and explain Air preheater with sketch. 4
- Q5(a) In a test of throttling calorimeter, the pressure before and after throttling are 11 bar and 1 bar respectively. Calculate least dryness fraction of steam. 4
- (b) Explain with neat sketch the working of Cochran boiler. 4
- (c) Give classification of boilers. 2

OR

- Q5(a) Explain Construction and working of Babcock and Wilcox boiler with neat sketch. 6
- (b) What is refrigeration? What is refrigerating effect? What is ton of refrigeration? 4
- Q6(a) Classify Air compressors. Explain working of two stage reciprocating Air compressor with neat sketch. 5
- (b) Give classification of Pumps. Explain Centrifugal pump with sketch. 5
- (c) Differentiate clutch, coupling and brakes. Draw line diagram of each. 5

OR

- Q6(a) Compare belt drive, chain drive and gear drive. Sketch different gears. 5
- (b) Explain with neat sketch working of single acting reciprocating pump. 5
- (c) Derive the equation of volumetric efficiency of a single stage reciprocating air compressor. 5